

# ShinyMR Promotes Reproducible and Rigorous Mendelian Randomization Studies with GWAS Summary Statistics

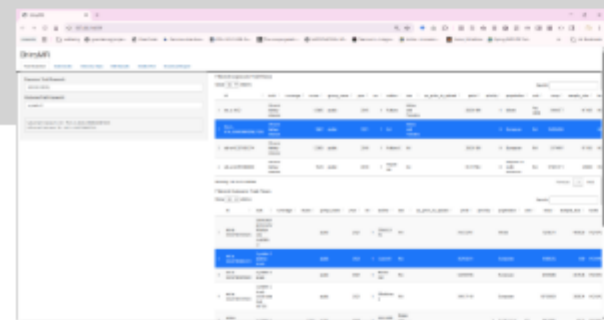
AUTHOR

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## Problem statement

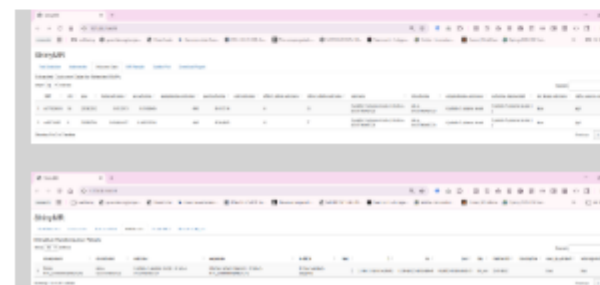
- Two-Sample MR is widely used with GWAS summary statistics like those in the OpenGWAS database and the GWAS Catalog (Davey Smith & Ebrahim, 2003; Hartwig et al., 2016).
- Ease of data access has fueled proliferation of MR studies that lack rigor and may provide misleading results (Elsworth et al., 2020; Lu et al., 2025; MacArthur et al., 2017).
- We developed ShinyMR (<https://github.com/fboehm/shinyMR>) to combat this problem.

## User interface



The screenshot shows the ShinyMR web application interface. It features a sidebar with navigation tabs (Home, Input, Output, Results) and a main content area displaying a table of GWAS summary statistics. The table has columns for SNP ID, P-value, R-squared, and other genetic data. The interface is designed for interactive data exploration and analysis.

ShinyMR provides a reproducible Rmarkdown file for primary MR analysis and sensitivity analyses to guide assessments of biases due to assumption violations



The two screenshots show different views of the ShinyMR web application. The top screenshot displays the 'Home' tab, which provides an overview of the application and its features. The bottom screenshot shows the 'Input' tab, where users can enter their GWAS summary statistics and configure the analysis parameters. Both views include a table of GWAS summary statistics with columns for SNP ID, P-value, R-squared, and other genetic data.

## References

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- Hartwig, F. P., Davies, N. M., Hemani, G., & Davey Smith, G. (2016). Two-sample mendelian randomization: Avoiding the downsides of a powerful, widely applicable but potentially fallible technique. In *International journal of epidemiology* (R; Vol. 45, pp. 1717–1725). Oxford University Press.
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<https://github.com/fboehm/shinyMR>